NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2023-2024

EE2005 – ELECTRICAL DEVICES AND MACHINES

November / December 2023 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Consider the magnetic circuit shown in Figure 1. The left limb is wound with a 500-turn coil, and there is also an air gap of length $l_g = 1$ mm in the left limb. The mean lengths of the limbs are indicated, namely, $l_{1a} = 34.9$ cm, $l_{1b} = 10$ cm, $l_2 = 15$ cm, and $l_3 = 45$ cm. The cross-sectional areas of the left, center and right limbs are 2.25 cm^2, 6 cm^2 and 2.25 cm^2, respectively. The relative permeability of the magnetic core is $\mu_r = 1500$ and the permeability of free space is $\mu_o = 4\pi x10^{-7}$ H/m. Ignore leakage and fringing effects.

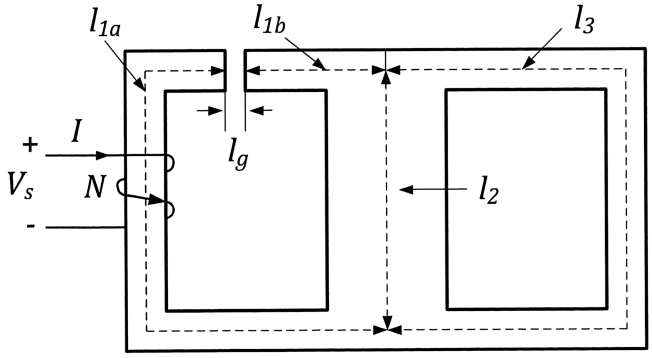


Figure 1

Note: Question No. 1 continues on page 2.

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Draw the magnetic equivalent circuit of the magnetic circuit shown in Figure 1 (on page 1).

(2 Marks)

(b) Suppose a flux density of $B_3 = 0.75$ T in the right limb is required. Determine:

(i) The individual fluxes in the left and center limbs.

(ii) The current in the coil to establish the required flux density in the right limb.

(iii) The energies stored in the air gap and core, respectively.

(iv) The new current needed to establish a flux density of $B_3 = 1$ T in the right limb, either using the values obtained in parts (i) and (ii) or otherwise.

(17 Marks)

(c) Suppose the coil has a resistance of 2 Ω and is connected to a 50 Hz AC source with an amplitude of 35 V. Determine the amplitudes of the current in the coil and the flux density established in the left limb.

(6 Marks)

2. A 10-kVA, 230/460-V, 50-Hz single-phase transformer has the following circuit parameters referred to the low-voltage side:

$$R_{e1} = 0.025\ \Omega, \quad X_{e1} = 0.18\ \Omega, \quad R_{c1} = 1400\ \Omega, \quad X_{m1} = 580\ \Omega$$

(a) Draw the approximate equivalent circuit of the transformer with the circuit parameters referred to the high-voltage side.

(5 Marks)

(b) Using the approximate equivalent circuit of the transformer obtained in part (a) and at 0.8 power factor lagging, with rated secondary terminal voltage, determine:

(i) The input current and power factor at full load.

(ii) The efficiency of the transformer at full load.

(iii) The maximum efficiency of the transformer. State any assumption made.

You may take the rated secondary voltage as the reference, i.e., $\vec{V}_2 = 460\angle 0^o$ V.

(14 Marks)

Note: Question No. 2 continues on page 3.

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(c) Determine the respective voltage regulations of the transformer at full load with 0.8 power factor lagging and 0.988 power factor leading. Comment on the performances of the transformer under these operating conditions. Explain how minimum voltage regulation can be achieved?

(6 Marks)

3. (a) Consider a three-phase, 50-Hz, 6-pole induction motor. The stator copper losses and the rotational losses of the motor are 800 W and 1000 W, respectively.

(i) If the motor is running at a speed of 950 rpm and with a load torque of 180 Nm, determine the power developed by the motor and the efficiency of the motor.

(ii) If the frequency of the voltage induced in the rotor is 2 Hz, determine the speed of the motor.

(7 Marks)

(b) A three-phase, 460-V, 60-Hz, 8-pole, wye-connected induction motor has the following per phase parameters referred to the stator:

$$R_1 = 0.1\,\Omega,\ X_M = 20\,\Omega,\ X_1 = 1\,\Omega,\ X_2 = 1\,\Omega,\ R_2 = \left(0.1 + R_{ext}\right)\,\Omega$$

where R_{ext} is the external resistance that is inserted in series in each phase of the rotor and referred to the stator per phase.

(i) If $R_{ext} = 0\,\Omega$, determine the input power taken by the motor from the supply when the motor develops its maximum torque and the value of the maximum torque that the motor can develop.

(ii) Determine the value of R_{ext} such that maximum torque is developed at starting.

(12 Marks)

(c) Consider a three-phase, 440-V, 50-Hz, 4-pole, wye-connected induction motor which has the following per phase parameters referred to the stator:

$$R_1 = 0\,\Omega,\ X_1 = 0\,\Omega,\ X_2 = 0.3\,\Omega,\ R_2 = 0.03\,\Omega, X_M = 28\,\Omega$$

If the motor is running at a speed of 1410 rpm at full load, determine the full-load torque developed.

(6 Marks)

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4. (a) The armature winding resistance and field winding resistance of a dc shunt motor are 0.1 Ω and 100 Ω, respectively. The armature terminals are connected to a 200-V dc supply. An external variable resistance R_{ext} is connected in series with the field winding and the rotational losses of the motor are 300 W.

(i) With $R_{ext} = 0\ \Omega$, the motor is running at a speed of 1200 rpm and drawing a line current of 200 A from the supply. Determine the torque developed, output power and efficiency of the motor.

(ii) With $R_{ext} = 0\ \Omega$, determine the speed of the motor if the motor is drawing a line current of 150 A from the supply.

(iii) Determine the value of R_{ext} for the motor to be running at a speed of 1500 rpm and drawing a line current of 200 A from the supply. Assume that the machine is operating in the linear part of the magnetization curve.

(15 Marks)

(b) A separately-excited dc generator has an armature winding resistance and a field winding resistance of 0.3 Ω and 120 Ω, respectively. The field circuit is connected to a 120-V dc supply. The machine is driven by an external prime mover and the effect of armature reaction can be neglected. The constant rotational losses of the machine are 500 W.

(i) When the speed of the external prime mover is 1000 rpm, the internal generated voltage is 100 V at no load. Determine the efficiency of the generator when it is delivering an armature current of 20 A to an external load.

(ii) Determine the speed of the external prime mover if the generator is delivering an armature current of 20 A to an external load at a terminal voltage of 114 V. Also determine the shaft torque of the external prime mover driving the generator. Assume that the field current is unchanged.

(10 Marks)

END OF PAPER

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EE2005 ELECTRICAL DEVICES & MACHINES

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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